Histometric indices for quantitative assessment of structural resistance of *Capsicum* spp. genotypes to *Meloidogyne enterolobii*

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Plant resistance to *Meloidogyne* spp. is conventionally assessed by the reproduction factor (RF), gall index, and egg mass index, variables that describe outcomes of parasitism without capturing the underlying cellular processes in the pathogen-host relationship. This study proposed three histometric indices for quantitative assessment of feeding site development in *Capsicum–M. enterolobii* interactions: Vascular Cylinder Area Index (VCAI), Giant Cell Occupation Index (GCOI), and Average Area of Giant Cells (AAGC). The indices were calculated from measurements on toluidine blue-stained histological root sections analyzed by digital morphometry software, in resistant and susceptible genotypes representing four *Capsicum* species inoculated with *M. enterolobii*. GCOI showed the highest discriminatory capacity among the evaluated genotypes, with large and consistent effects across all evaluated species ($d = 1.10–1.84$), although the direction of the response varied according to the taxon. Integration of histometric indices with RF allowed identification and validation of resistance and susceptibility patterns. The resistant genotype *C. chinense* UFGMH005 exhibited the greatest suppression of giant cell development, associated with preservation of vascular organization, a pattern consistent with an early resistance mechanism and confirmed by RF < 1.0. The opposite result was observed in the susceptible genotype of the same species, UFGMH049. No significant correlations were detected between individual indices and RF across species, indicating that histometric responses reflect species-specific structural resistance patterns. The proposed indices constitute reproducible quantitative descriptors of feeding site development, with potential application in breeding programs targeting *Meloidogyne* spp. management and other crops.

Keywords: Feeding Site, Giant Cells, Histometry, Histopathology, Resistance Phenotyping.